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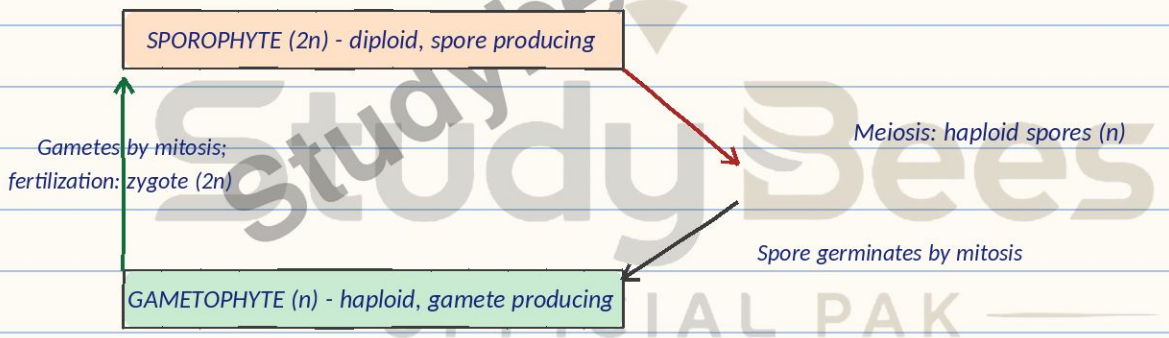
# Chapter 8: Kingdom Plantae

## 8.1 Overview of Kingdom Plantae

- Plants began in water; migration to land started nearly **400 million years ago**. Freshwater green algae **Charophyceae** (e.g. **Chara**) are the closest relatives of land plants (common ancestor, monophyletic lineage; not ancestors): cellulose-rich walls, similar flagellated sperm, similar nuclear and chloroplast genes.

### Diagnostic features of plants

- Multicellular eukaryotes with tissues; autotrophic; **cuticle of cutin** prevents drying; body of root, stem, leaves with **xylem and phloem**; cellulose walls; **alternation of generations**; multicellular sex organs; **oogamous** gametes (different).




Heteromorphic alternation: the two generations look different

Fig: Alternation of generations in plants

**Sporophyte:** The diploid generation that produces haploid spores by meiosis.

**Gametophyte:** The haploid generation with multicellular sex organs producing gametes (by mitosis) that unite to form a diploid zygote.

- **Heteromorphic alternation of generations:** sporophyte and gametophyte differ morphologically.

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## Classification (SLO 4)

- Phylogenetically two groups: **Bryophytes** (non-vascular: liverworts, mosses, hornworts) and **Tracheophytes** (vascular): **Lycophytes** and **Euphyllophytes** (ferns, horsetails, seed plants). Lycophytes, ferns and horsetails = **pteridophytes** (free sporing, paraphyletic); **spermatophytes** (seed plants, monophyletic) = gymnosperms (non-flowering) + angiosperms (flowering).

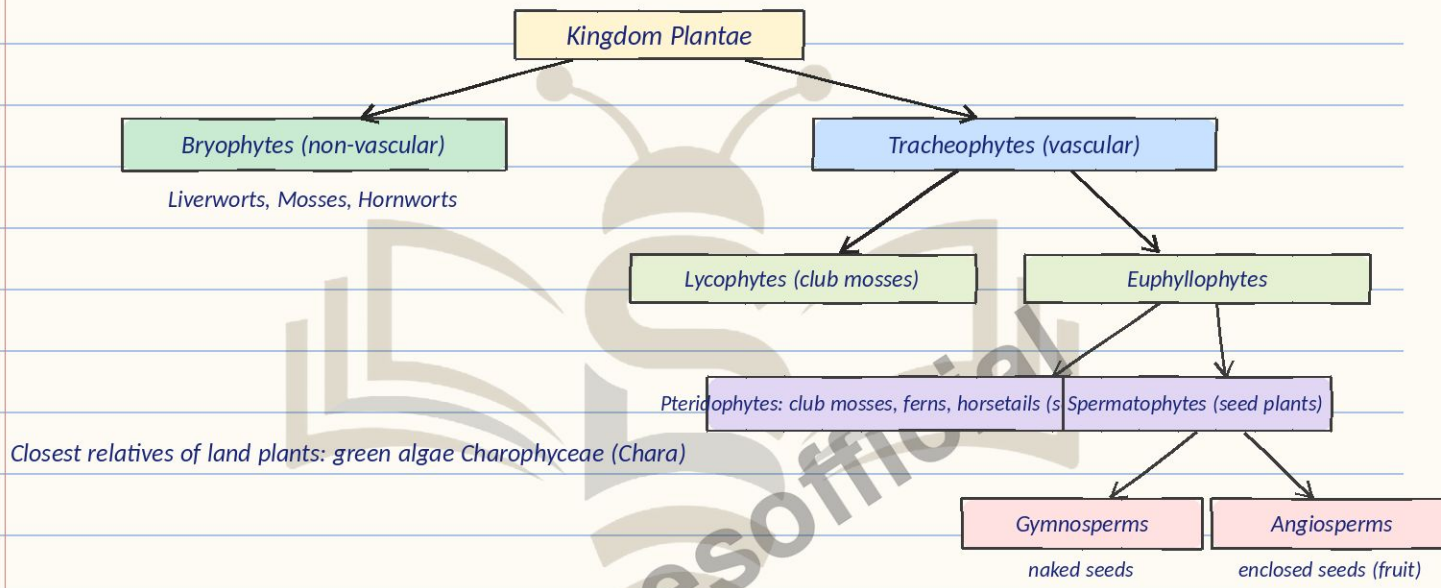



Fig: Phylogeny of Kingdom Plantae

## 8.2 Bryophyta - non-vascular plants (SLO 1, 2)

- Liverworts, hornworts, mosses: no vascular tissue, no true roots, stems, leaves (root-, stem-, leaf-like structures); small (rarely over 2 cm); live mostly in wet places; **swimming male gametes need water** (hence "amphibious plants").
- Main body is the **gametophyte** (green, larger, conspicuous). **Sporophyte** is smaller, obtains food from the gametophyte, homosporous.
- **Four main features:** (1) no vascular tissues; (2) multicellular sex organs produce embryo; (3) sporophyte smaller and dependent on gametophyte; (4) sporophyte homosporous.

## Land adaptations of bryophytes

1. Compact multicellular body reduces water loss; cuticle.
2. Photosynthetic tissue in special air chambers: $CO_2$ absorption with minimum

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*water loss.*

3. ***Rhizoids** for water absorption and anchorage.*
4. *Heterogamy (oogamy): non-motile egg with stored food, motile sperm.*
5. *Multicellular **antheridia and archegonia** protect gametes.*
6. *Multicellular embryos retained and protected in female organ.*
7. *Alternation of generations gives genetic variety.*

- ***Subdivisions:** Hepaticopsida (liverworts, Marchantia), Bryopsida (mosses, Polytrichum, Funaria), Anthoceropsida (hornworts). Importance: hold soil (prevent erosion), food for birds and small mammals; peat moss holds water, a soil conditioner.*

## *Life cycle of Polytrichum (moss)*

- ***Gametophyte:** leafy erect shoot anchored by rhizoids; sex organs at apex. **Antheridium:** multicellular stalk and capsule with sterile jacket; produces many **biflagellate sperms** by mitosis. **Archegonium:** swollen **venter** (one egg) and long **neck**; sterile jacket. Organs mixed with hair-like **paraphyses**. Polytrichum is **dioecious** (separate male and female plants); **Funaria is monoecious**.*
- ***Fertilization:** sperms swim in a film of water through mucilaginous passage of the neck to the egg: diploid **zygote** in archegonium.*
- ***Sporophyte:** zygote to embryo to mature sporophyte: **foot** (embedded in gametophyte), **seta** (stalk) and **capsule** (sporangium) with spores by meiosis. Spore (n, first cell of gametophyte) germinates into alga-like **protonema** with buds that grow into the gametophyte.*

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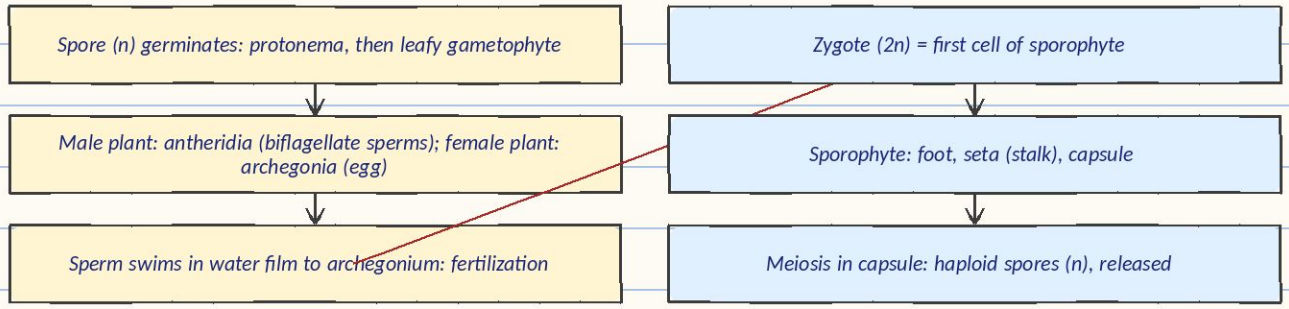



*Fig: Life cycle of Polytrichum (moss); dioecious, gametophyte dominant*

## 8.3 Tracheophyta - vascular plants (SLO 3, 4)

- *Have **xylem and phloem**; named for **tracheids** in xylem. Success due to: (1) sporophyte main body with roots, stems, leaves and vascular system; (2) protected sporangia leading to seeds; (3) **pollen tube** (water-independent transfer of male gametes); (4) flowers and fruits; (5) heteromorphic alternation of generations.*
- *Four subdivisions: **Psilopsida, Lycopsida, Sphenopsida, Pteropsida**.*

## 8.4 Pteridophyta - seedless vascular plants (SLO 5)

| Class | Characteristics and examples |
|---|---|
| Psilopsida (whisk ferns) | Simplest vascular plants; no true roots (underground rhizomes with rhizoids); aerial stems without true leaves (small green scales); sporangia at tips / sides; gametophyte colourless, underground with fungus. Rhynia (extinct), Psilotum. |
| Lycopsida (club mosses) | Branching rhizome, aerial stems under 30 cm; scale-like **microphylls** with a single vascular strand (only living plants with microphylls); sporangia on **strobili**; underground gametophyte. Lycopodium, Selaginella. |

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| Class | Characteristics and examples |
|---|---|
| Sphenopsida (horsetails) | Only living genus **Equisetum**; rhizome gives fertile and sterile aerial stems (green, hollow, jointed = arthrophytes); whorls of leaves at nodes; spore rings in cones; thalloid gametophyte on mud. |
| Pteropsida: Fillicinae (ferns) | Most abundant seedless vascular plants in warm moist tropics; size under 1 cm to 24 m tree ferns with 5 m leaves; mostly homosporous; sporophyte dominant and independent; sporangia on fronds; young fronds coiled = **circinate vernation**. |

- ***Importance:*** *Lycopodium and Selaginella ornamental, Christmas wreaths; fern fronds in bouquets; tree fern stems for building; Pteris, Ceratopteris, Marsilea edible; male fern rhizome drug removes intestinal parasites; seedless vascular plants formed coal.*

## *Life cycle of fern - Adiantum (SLO 6)*

- ***Sporophyte:*** *underground horizontal rhizome with brown scales (**ramenta**), fibrous adventitious roots, large pinnately compound fronds (young = **fiddleheads**, circinate vernation); dark shiny **rachis** (maidenhair); pinnae and pinnules with dichotomous venation. **Sori** (clusters of sporangia) under the margins, covered by bent margin = false indusium.*
- ***Sporangium:*** *stalk + capsule; wall with **annulus** (thick cells) and **stomium** (thin cells, bursting site). Spore mother cells divide by meiosis to many spores (exine and intine). On drying annulus contracts and bursts the stomium: spores dispersed. Spore germinates: exine bursts, intine grows into tube giving the gametophyte.*
- ***Gametophyte (prothallus):*** *independent, green, **heart shaped** with central notch (growing point), rhizoids below. **Archegonia** flask shaped (venter with egg, neck canal cells); **antheridia** globular, with spiral multiciliate antherozoids needing water.*
- ***Fertilization:*** *antherozoids swim by chemotaxis; zygote (oospore) develops to*

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*embryo then young sporophyte, first dependent on gametophyte, soon independent. Heteromorphic alternation.*

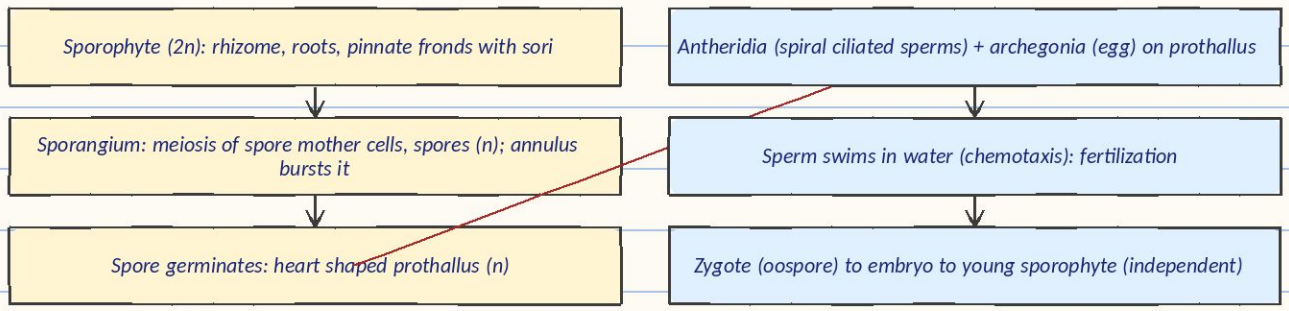



*Fig: Life cycle of fern (Adiantum)*

## 8.5 Spermatophyta - seed plants (SLO 7)

- *Gymnosperms (exposed seeds on sporophylls in cones) and Angiosperms (seeds enclosed in fruits).* ***Common adaptations:***

1. ***Heterospory:*** *microspores (male) and megaspores (female).*
2. *Indehiscent megasporangium (**ovule**): megaspore retained and germinates inside.*
3. ***Integuments*** *protect the megasporangium.*
4. *Only* ***one functional megaspore*** *of four survives (less competition).*
5. ***Micropyle*** *opening at distal end for pollen tube.*
6. *Narrow* ***pollen tube*** *for safe sperm transport (water-independent).*
7. *Zygote to embryo; whole ovule becomes* ***seed****; integuments form the* ***seed coat****.*

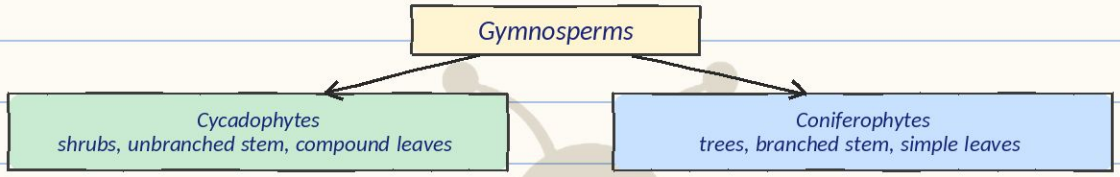
### 8.5.1 Gymnosperms (naked seeded)

- *About one third of world forests; tall woody perennial trees or shrubs; Gk gymnos = naked, sperma = seed: ovules exposed on megasporophylls. Heterosporous;* ***seeds but no fruits****.*
- ***Xerophytic leaves:*** *thick tough cuticle, stomata sunken in pits, waxy surface.*

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***Xylem:*** *tracheids and xylem parenchyma (no vessels or fibres except Gnetales);* ***phloem: no companion cells****. Dominant sporophyte, dependent gametophyte. Microsporophylls / megasporophylls spirally arranged in male / female* ***cones (strobili)****.* ***Polyembryony*** *is common; finally one embryo matures.*

- *Genera:* ***Cycas*** *(sago palm),* ***Pinus****,* ***Taxus*** *(yew),* ***Picea*** *(hemlock),* ***Cedrus*** *(deodar),* ***Ginkgo****, Ephedra.*




*Orders: Cycadofilicales (extinct), Bennettitales (extinct), Cycadales (living)Cordaitales (extinct), Ginkgoales, Coniferales, Gnetales (living)*

*Genera: Cycas (sago palm), Pinus, Taxus (yew), Picea, Cedrus (deodar), Ginkgo*

*Fig: Classification of gymnosperms*

- ***Importance:*** *Ephedra drug for asthma; conifers: softwood for construction, packing, plywood, paper; cycads ornamental, Cycas circinalis source of sago (pure starch); resins, turpentine, tar and oils from conifers.*

## *Life cycle of Pinus (SLO 8)*

- ***Sporophyte:*** *tall woody trunk, spiral branches, evergreen needle-like leaves (water conserving), taproot system. Cones:* ***male cone*** *(staminate strobilus: small, clusters on upper branches, microsporophylls with 2 microsporangia each, many microspores by meiosis);* ***female cone*** *(ovulate strobilus: 6 to 12 inches, lower branches, woody scales = megasporophylls with 2 ovules each; megaspore mother cell gives 4 megaspores, one functional).*
- ***Pollination:*** *winged microspores (pollen grains) carried by wind to the micropyle; here pollen grains are developing male gametophytes (****endosporic****).*
- ***Male gametophyte:*** *microspore nucleus divides to prothallial and antheridial nuclei; prothallial divides again (2 vegetative nuclei); antheridial divides to generative and tube nucleus; generative gives 2 sperm nuclei; pollen tube by tube nucleus.*
- ***Female gametophyte:*** *functional megaspore divides by mitosis to multicellular*

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*embryo sac with archegonia, each with an egg.*

- ***Fertilization:*** *pollen tubes enter through micropyle and fuse with archegonial neck; several eggs fertilized (**polyembryony**) but usually one embryo survives; each male gametophyte gives one sperm = **single fertilization**. Ovule becomes seed with embryo and stored food; seed germinates into a new sporophyte.*

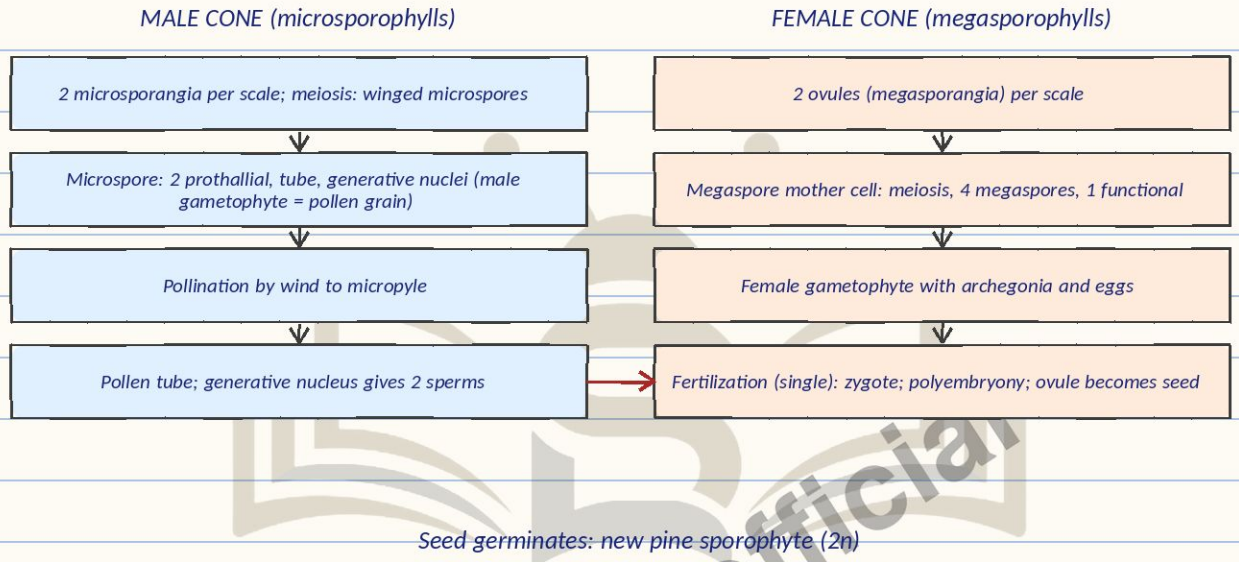



*Fig: Life cycle of Pinus (gymnosperm)*

## *8.5.2 Angiosperms (SLO 9, 10)*

- *Flowering plants; Gk angio = closed, sperm = seed. Seeds enclosed in **fruit** (ovary develops into fruit). Xylem has vessels and fibres; phloem has companion cells. Dominant sporophyte, dependent gametophyte.*

### *Sporophyte and flower*

- *Herb, shrub or tree. **Flower** = modified shoot: **pedicel**, **thalamus (torus)**, whorls: **calyx** (sepals), **corolla** (petals), **androecium** (stamens), **gynoecium** (carpels). Calyx and corolla = **accessory whorls** (protect, attract insects); androecium and gynoecium = **necessary whorls** (reproductive).*

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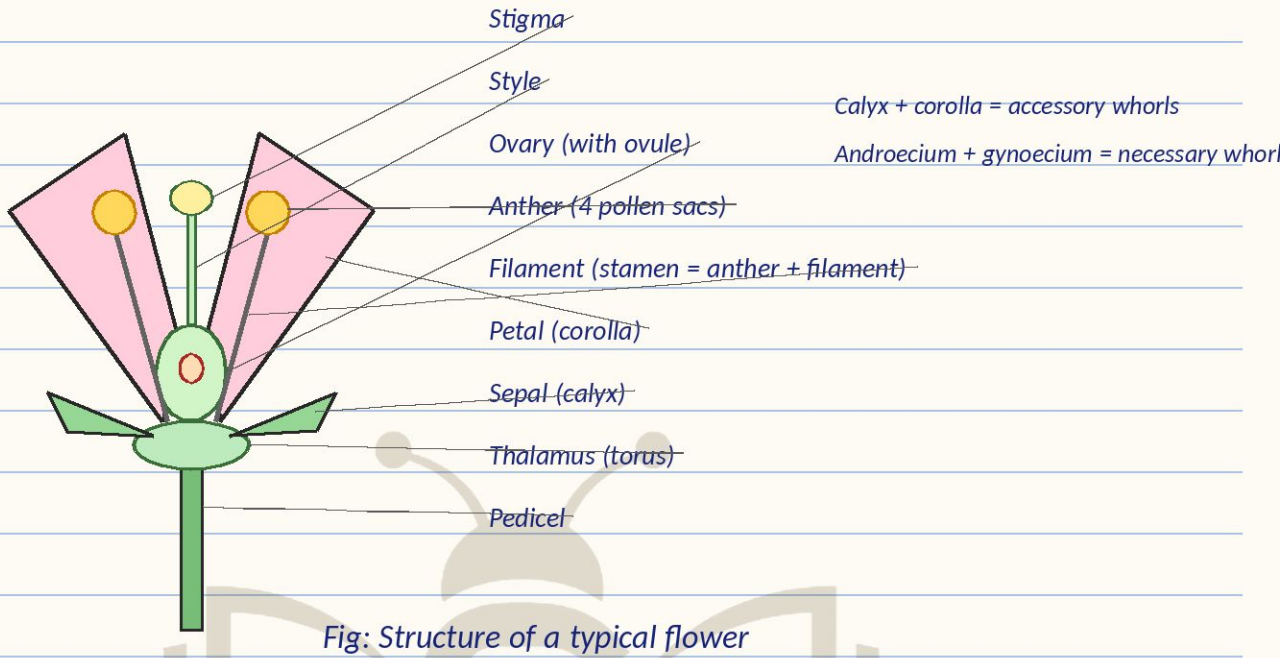



*Fig: Structure of a typical flower*

- ***Stamen:*** *filament + bilobed anther with 4 pollen sacs (microsporangia) producing haploid pollen grains (microspores) by meiosis.* ***Carpel (pistil):*** *ovary, style, stigma; ovary contains ovules (integumented indehiscent megasporangia); spore mother cell in each ovule gives 4 megaspores, one functional.*
- ***Pollination:*** *transfer of pollen from anther to stigma by wind, insects or animals. In angiosperms pollen grains germinate on the stigma, so the male gametophyte is* ***exosporic****.*
- ***Male gametophyte:*** *microspore nucleus gives tube nucleus + generative nucleus; generative gives* ***2 sperm nuclei****; tube nucleus forms the pollen tube down the style.*
- ***Female gametophyte (embryo sac):*** *functional megaspore nucleus divides by 3 mitoses to 8 nuclei: 3 at micropylar end (1* ***egg*** *+ 2* ***synergids****), 3 at opposite end (****antipodals****), 2* ***polar nuclei*** *in centre that may fuse to a diploid* ***secondary nucleus****.*

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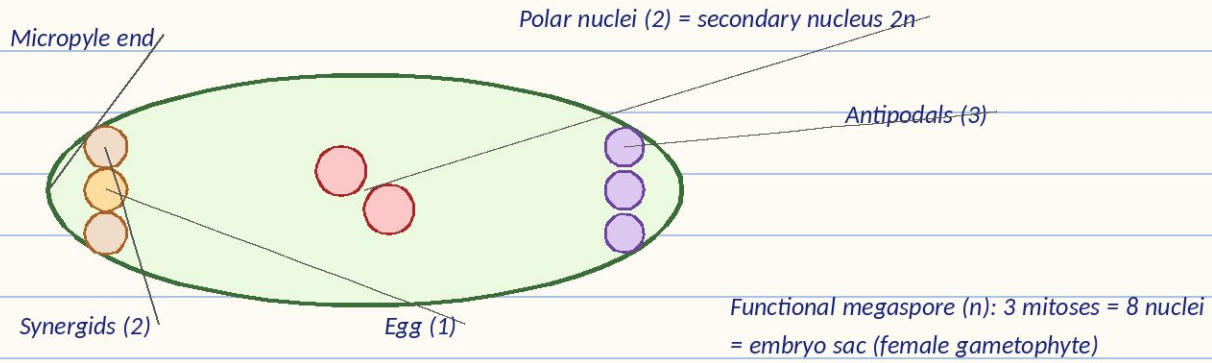
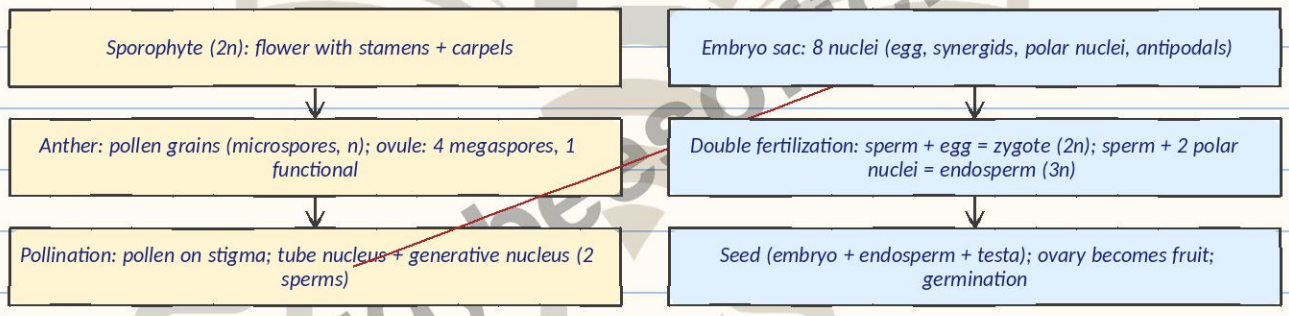



*Fig: Embryo sac of an angiosperm (7 celled, 8 nucleate)*

- ***Double fertilization:*** *one sperm + egg = diploid zygote (oospore); other sperm + polar nuclei / secondary nucleus = triploid* ***endosperm nucleus****. One male gametophyte contributes two sperm nuclei.*
- ***Seed formation:*** *zygote becomes embryo; endosperm nucleus becomes nutritive endosperm; integuments become seed coat (****testa and tegmen****); ovary wall becomes fruit. Dormancy, then germination: seedling to sporophyte.*




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*Fig: Life cycle of an angiosperm*

## *Classification of angiosperms: dicots and monocots*

| ***Dicotyledonae (dicots)*** | ***Monocotyledonae (monocots)*** |
|---|---|
| *Embryo with two cotyledons* | *Embryo with one cotyledon* |
| *Leaves with reticulate venation* | *Long narrow, lanceolate leaves, parallel venation* |

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| Dicotyledonae (dicots) | Monocotyledonae (monocots) |
|---|---|
| Vascular bundles in a ring; cambium present: secondary growth | Vascular bundles scattered; cambium usually absent: no secondary growth |
| Primary root is a tap root with lateral roots | Adventitious fibrous root system from stem base |
| Root: 2-8 patches of xylem, cambium present | Root: more than 8 xylem patches, no cambium |
| Floral parts in fours or fives (tetra / pentamerous); calyx and corolla differ; usually insect pollinated | Floral parts in threes (trimerous); no distinction between calyx and corolla; often wind pollinated |
| Rose, pea, buttercup | Lilies, orchids, grasses, wheat, rice |

## 8.5.4 Angiosperm families (SLO 11)

### Brassicaceae (Cruciferae) - mustard / cabbage family (dicot)

- **Habit:** mesophytes; mostly herbs with pungent smell. **Stem:** erect, herbaceous, solid, cylindrical, often hairy (**trichomes**). **Leaves:** simple, alternate. **Flowers:** usually bisexual; **four petals in a cross** (cruciform). **Fruits:** capsule or **siliqua**. **Chemicals:** glucosinolates (pungent taste).
- **Economic importance:** vegetables (Raphanus sativus radish, Brassica rapa, B. juncea, Lepidium sativum); **oil** from Brassica campestris, B. juncea (rayo), B. nigra, B. alba; seed cake as cattle feed and manure; fodder; condiments (mustard); medicinal (Iberis amara, Cheiranthus cheiri for asthma and bronchitis; Matthiola incana anti-inflammatory; Lepidium root); weeds of medicinal value (Capsella bursa-pastoris); ornamentals (candytuft, honesty, wallflower).

### Solanaceae - nightshade / potato family (dicot)

- **Habit:** annual or perennial herbs, shrubs, woody plants, deserts to rainforests. **Stem:** hairy or prickly, branched. **Leaves:** alternate (sometimes opposite), simple to pinnately compound, unpleasant odour. **Flowers:** pentamerous: 5 sepals, 5 united petals, tubular or funnel shaped, showy, fragrant. **Fruit:** berry

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*or capsule.* ***Chemicals: alkaloids*** *(nicotine in tobacco, tropane alkaloids in deadly nightshade).*

- ***Economic importance:*** *food:* ***Solanum tuberosum*** *(potato),* ***Lycopersicon esculentum*** *(tomato),* ***Solanum melongena*** *(brinjal),* ***Capsicum annuum / C. frutescens*** *(chillies; vitamins C and A, condiment),* ***Physalis*** *(ground cherry, husk tomato);* ***Nicotiana tabacum*** *(tobacco); drugs: Atropa belladonna (atropine), Datura (daturine); ornamentals: Petunia, Nicotiana, Cestrum, Solanum.*

## *Poaceae (Gramineae) - grass family (monocot)*

- ***Habit:*** *herbaceous, grass-like, worldwide (grasslands, savannas, wetlands).* ***Stem:*** *hollow or solid* ***culm****, round.* ***Leaves:*** *alternate, linear, parallel venation.* ***Flowers:*** *small, no showy petals, stamens and pistils;* ***wind pollinated****.* ***Fruit: caryopsis*** *(grain): one seeded, dry, indehiscent.*
- ***Economic importance:*** *staples: wheat (Triticum), oats (Avena sativa), maize (Zea mays), rice (Oryza sativa), barley (Hordeum vulgare), rye (Secale cereale), Pennisetum, Sorghum; fodder; sugar from* ***Saccharum officinarum*** *(sugarcane); lawn grasses (Agrostis, Poa, Festuca);* ***bamboo*** *(Bambusa) for building, mats, baskets; lemon grass (Cymbopogon citratus) oil for perfumes and soaps; paper; alcohol and beverages (whisky, rum); fibres of Saccharum munja for ropes.*

| Character | Brassicaceae | Solanaceae | Poaceae |
|---|---|---|---|
| Class | Dicot | Dicot | Monocot |
| Flower | 4 petals, cruciform | 5 sepals, 5 united petals | Reduced, wind pollinated |
| Fruit | Siliqua / capsule | Berry / capsule | Caryopsis (grain) |
| Chemicals | Glucosinolates | Alkaloids | Essential oils |
| Key uses | Oils, vegetables | Potato, tomato, tobacco | Cereals, sugar, bamboo |

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## *STEAM activity: Botanical detective*

- *Observe plant morphology outdoors with hand lenses, collect specimens, keep field notes with accurate sketches, and use identification guides or digital resources to identify plants and their families.*

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## *Chapter 8: Short Questions with Answers (SLO based)*

***Q1. [SLO 1] Which algae are the closest relatives of land plants? Why?***

*Ans: Charophyceae (Chara); they share cellulose rich walls, similar flagellated sperm and similar nuclear and chloroplast genes, and both come from one common ancestor.*

***Q2. [SLO 1] When did plants migrate to land?***

*Ans: Nearly 400 million years ago.*

***Q3. [SLO 1] List the diagnostic features of plants.***

*Ans: Multicellular eukaryotes with tissues, autotrophic, cuticle of cutin, root-stem-leaf body with xylem and phloem, cellulose walls, alternation of generations, oogamy.*

***Q4. [SLO 1] Define sporophyte and gametophyte.***

*Ans: Sporophyte: diploid generation producing haploid spores by meiosis. Gametophyte: haploid generation producing gametes by mitosis.*

***Q5. [SLO 1] What is heteromorphic alternation of generations?***

*Ans: Alternation of sporophyte and gametophyte generations that differ in morphology.*

***Q6. [SLO 4] Divide the plant kingdom phylogenetically.***

*Ans: Bryophytes (non-vascular) and Tracheophytes (vascular: lycophytes and euphyllophytes); seed plants divide into gymnosperms and angiosperms.*

***Q7. [SLO 4] What are pteridophytes?***

*Ans: Free-sporing seedless vascular plants: lycophytes, ferns and horsetails.*

***Q8. [SLO 1] Why are bryophytes called amphibious plants?***

*Ans: They live mostly in wet places and need water for their swimming male gametes to reach the egg.*

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**Q9. [SLO 1] What is the main plant body of bryophytes?**

*Ans: The green, conspicuous gametophyte; the sporophyte is small and depends on it.*

**Q10. [SLO 1] List the four main features of bryophytes.**

*Ans: No vascular tissue; multicellular sex organs producing embryo; smaller sporophyte getting food from gametophyte; sporophyte homosporous.*

**Q11. [SLO 1] What are land adaptations of bryophytes? Name three.**

*Ans: Rhizoids for anchorage, cuticle and compact body, air chambers for gas exchange, antheridia and archegonia protecting gametes, retained embryo.*

**Q12. [SLO 1] Name the three subdivisions of bryophytes.**

*Ans: Hepaticopsida (liverworts), Bryopsida (mosses), Anthoceropsida (hornworts).*

**Q13. [SLO 1] What is peat moss used for?**

*Ans: Its leaves hold water, so it is a soil conditioner that helps sandy soil retain moisture.*

**Q14. [SLO 2] Differentiate between antheridium and archegonium.**

*Ans: Antheridium: male organ with stalk and capsule producing biflagellate sperms. Archegonium: female flask-shaped organ with venter (egg) and neck.*

**Q15. [SLO 2] What are paraphyses and rhizoids?**

*Ans: Paraphyses: multicellular hair-like structures among sex organs. Rhizoids: root-like structures for anchorage and absorption.*

**Q16. [SLO 2] Differentiate between dioecious and monoecious.**

*Ans: Dioecious: antheridia and archegonia on separate plants (Polytrichum). Monoecious: both on the same plant (Funaria).*

**Q17. [SLO 2] What is protonema?**

*Ans: The alga-like filamentous structure formed when a moss spore germinates; its buds grow into the gametophyte.*

**Q18. [SLO 2] Name the parts of the moss sporophyte.**

*Ans: Foot, seta (stalk) and capsule (sporangium).*

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**Q19. [SLO 3] Why are tracheophytes so named?**

*Ans: They have tracheids in the xylem; they have xylem and phloem.*

**Q20. [SLO 3] List the general characteristics of vascular plants.**

*Ans: Dominant sporophyte with roots, stems, leaves and vascular tissue; protected sporangia / seeds; pollen tube; flowers and fruits; heteromorphic alternation.*

**Q21. [SLO 4] Name the four subdivisions of tracheophytes.**

*Ans: Psilopsida, Lycopsida, Sphenopsida and Pteropsida.*

**Q22. [SLO 5] Why are whisk ferns considered the simplest vascular plants?**

*Ans: They have no true roots (rhizomes with rhizoids) and no true leaves (small green scales).*

**Q23. [SLO 5] What are microphylls and strobili?**

*Ans: Microphylls: small scale-like leaves with a single vascular strand (club mosses). Strobili: club-shaped clusters of leaves bearing sporangia.*

**Q24. [SLO 5] What are arthrophytes? Name the genus.**

*Ans: Horsetails with green hollow jointed stems; only living genus Equisetum.*

**Q25. [SLO 5] What is circinate vernation?**

*Ans: The coiled young fern frond which uncoils into a flat leaf (fiddlehead).*

**Q26. [SLO 5] Write four features of ferns.**

*Ans: Mostly homosporous; dominant independent sporophyte; sporangia on fronds; circinate vernation.*

**Q27. [SLO 5] Write the importance of seedless vascular plants.**

*Ans: Ornamentals, edible ferns (Pteris, Marsilea), male fern drug against intestinal parasites, building material, coal formation.*

**Q28. [SLO 6] What are sori, annulus and stomium?**

*Ans: Sori: clusters of sporangia. Annulus: thick-walled cell portion of sporangium wall. Stomium: thin-walled cells where the sporangium bursts.*

**Q29. [SLO 6] Describe the gametophyte of Adiantum.**

*Ans: Independent, green, heart shaped prothallus with rhizoids; bears flask shaped archegonia and globular antheridia.*

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**Q30. [SLO 6] Why do ferns need water for fertilization?**

*Ans: The antherozoids are multiciliate and swim in water toward the archegonia.*

**Q31. [SLO 7] List the common adaptations of seed plants.**

*Ans: Heterospory, indehiscent ovule, integuments, one functional megaspore, micropyle, pollen tube and seed formation.*

**Q32. [SLO 7] What are micropyle and integuments?**

*Ans: Micropyle: opening at the distal end of ovule for the pollen tube. Integuments: protective layers around the megasporangium that form the seed coat.*

**Q33. [SLO 7] Define gymnosperm. Give examples.**

*Ans: Naked seeded plants whose ovules are exposed on megasporophylls: Pinus, Cycas, Taxus, Cedrus, Ginkgo, Picea.*

**Q34. [SLO 7] What are the xerophytic features of gymnosperm leaves?**

*Ans: Thick tough cuticle, sunken stomata in pits and wax on the surface.*

**Q35. [SLO 7] How does gymnosperm xylem and phloem differ from angiosperm?**

*Ans: Gymnosperm xylem has tracheids (no vessels and fibres except Gnetales) and phloem has no companion cells.*

**Q36. [SLO 7] What is polyembryony?**

*Ans: Fertilization of several eggs giving several embryos in the same embryo sac; common in gymnosperms; usually one survives.*

**Q37. [SLO 7] Classify gymnosperms.**

*Ans: Cycadophytes (shrubs, unbranched stem, compound leaves) and Coniferophytes (trees, branched stem, simple leaves).*

**Q38. [SLO 7] Write the importance of gymnosperms.**

*Ans: Ephedra for asthma; conifers give softwood and paper; Cycas gives sago; resins, turpentine and tar.*

**Q39. [SLO 8] Differentiate between male and female cones of Pinus.**

*Ans: Male cones: small, clusters on upper branches, microsporophylls with microsporangia. Female cones: 6-12 inches, lower branches, woody megasporophylls with 2 ovules each.*

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***Q40. [SLO 8] What is single fertilization?***

*Ans: Each male gametophyte of gymnosperm contributes only one sperm for fertilization.*

***Q41. [SLO 8] What is meant by endosporic male gametophyte?***

*Ans: Microspores begin to germinate into male gametophytes inside the microsporangium (pollen grain = developing male gametophyte in Pinus).*

***Q42. [SLO 9] Define angiosperms. Why are they called so?***

*Ans: Flowering plants with seeds enclosed in fruits (Gk angio = closed, sperm = seed).*

***Q43. [SLO 9] Differentiate between accessory and necessary whorls of a flower.***

*Ans: Accessory: calyx and corolla (protect and attract). Necessary: androecium and gynoecium (reproductive).*

***Q44. [SLO 9] Name the parts of a stamen and a carpel.***

*Ans: Stamen: filament and anther. Carpel: ovary, style and stigma.*

***Q45. [SLO 10] What is an embryo sac?***

*Ans: The female gametophyte of angiosperm with 8 nuclei: egg, 2 synergids, 3 antipodals and 2 polar nuclei.*

***Q46. [SLO 10] What is double fertilization?***

*Ans: One sperm fuses with the egg (zygote) and the other with polar nuclei (triploid endosperm nucleus).*

***Q47. [SLO 10] What does an ovule become after fertilization? What does the ovary become?***

*Ans: The ovule becomes the seed (integuments form testa and tegmen) and the ovary wall becomes the fruit.*

***Q48. [SLO 10] What is the male gametophyte of angiosperms?***

*Ans: Exosporic: pollen grain germinates on the stigma; the structure with tube nucleus and two sperm nuclei is the male gametophyte.*

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**Q49. [SLO 10] Differentiate between dicots and monocots (any four).**

Ans: Dicots: two cotyledons, reticulate venation, tap root, vascular bundles in ring with cambium. Monocots: one cotyledon, parallel venation, fibrous roots, scattered bundles without cambium.

**Q50. [SLO 11] Why is Brassicaceae also called Cruciferae?**

Ans: Its flowers have four petals in a cross-like arrangement (cruciform).

**Q51. [SLO 11] What is the fruit of Brassicaceae? What gives the pungent taste?**

Ans: Siliqua or capsule; glucosinolates.

**Q52. [SLO 11] Give economic uses of Brassicaceae.**

Ans: Vegetables (radish), oils from mustard (Brassica), fodder, condiments, medicines, ornamentals.

**Q53. [SLO 11] What compounds are found in Solanaceae? Give examples.**

Ans: Alkaloids: nicotine in tobacco, atropine in Atropa belladonna, daturine in Datura.

**Q54. [SLO 11] Write the floral characters of Solanaceae.**

Ans: Pentamerous: five sepals and five united petals, tubular or funnel shaped, showy and fragrant.

**Q55. [SLO 11] Name important food plants of Solanaceae.**

Ans: Potato, tomato, brinjal (egg plant), chillies (Capsicum), ground cherry.

**Q56. [SLO 11] Why is Poaceae the most economically important monocot family?**

Ans: It provides staple cereals (wheat, rice, maize, barley, oats), sugarcane, fodder, bamboo and many other products.

**Q57. [SLO 11] What is a caryopsis? What is a culm?**

Ans: Caryopsis: one-seeded dry indehiscent grain fruit of grasses. Culm: stem of grasses (hollow or solid).

**Q58. [SLO 11] Why are grass flowers not showy?**

Ans: They are wind pollinated and do not need colourful petals to attract pollinators.

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# Chapter 8: Long Questions with Answers (SLO based)

## Q1. [SLO 1] Describe the diagnostic features of plants and the alternation of generations.

**Ans:**

- Multicellular eukaryotes with tissues; autotrophic; cuticle; root, stem, leaves with xylem and phloem; cellulose walls; oogamous gametes with multicellular sex organs.

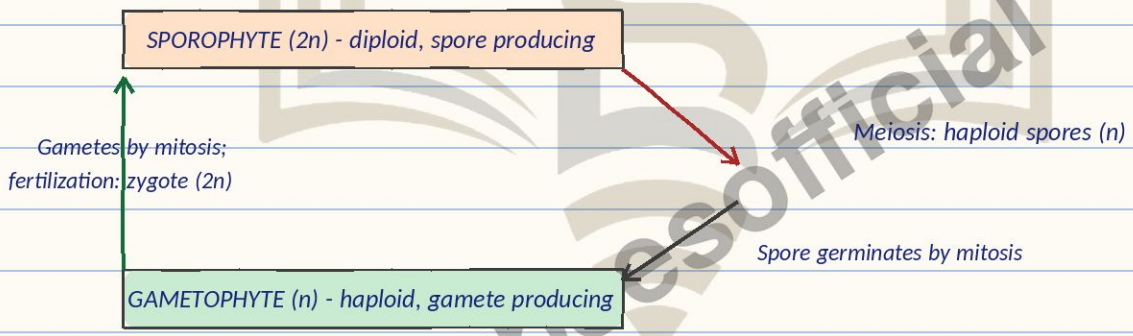



Fig: Alternation of generations in plants

- Sporophyte (2n) produces spores (n) by meiosis; spores develop into gametophyte (n); gametophyte makes gametes by mitosis; fusion gives zygote (2n) that grows into the sporophyte. Heteromorphic: generations look different.

## Q2. [SLO 1] Describe non-vascular plants (bryophytes) with their land adaptations.

**Ans:**

- Liverworts, mosses, hornworts: no vascular tissue; no true roots, stems, leaves; small; moist habitats; swimming sperm; dominant gametophyte; smaller dependent homosporous sporophyte.

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- *Adaptations: compact body and cuticle, air chambers, rhizoids, oogamy, antheridia and archegonia, protected embryo, alternation of generations.*
- *Subdivisions: Hepaticopsida, Bryopsida, Anthoceropsida. Importance: soil binding, food for animals, peat moss.*

**Q3. [SLO 2] Explain the life cycle of Polytrichum.**

**Ans:**

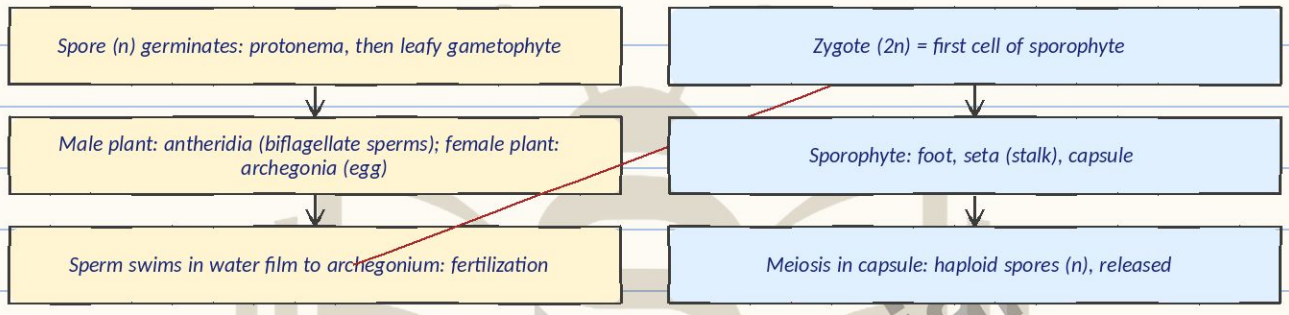



*Fig: Life cycle of Polytrichum (moss); dioecious, gametophyte dominant*

- *Gametophyte: leafy shoot with rhizoids; antheridia (biflagellate sperms) and archegonia (venter, egg, neck) on separate plants (dioecious) with paraphyses. Sperm swims through a water film to the egg.*
- *Sporophyte: zygote to embryo to foot, seta, capsule; meiosis gives haploid spores; spore germinates to protonema then gametophyte.*

**Q4. [SLO 3, 4] Describe the general features of vascular plants and their division into pteridophytes, gymnosperms and angiosperms.**

**Ans:**

- *Vascular plants: xylem and phloem; dominant sporophyte with roots, stems, leaves; protected sporangia leading to seeds; pollen tube; flowers and fruits; heteromorphic alternation.*

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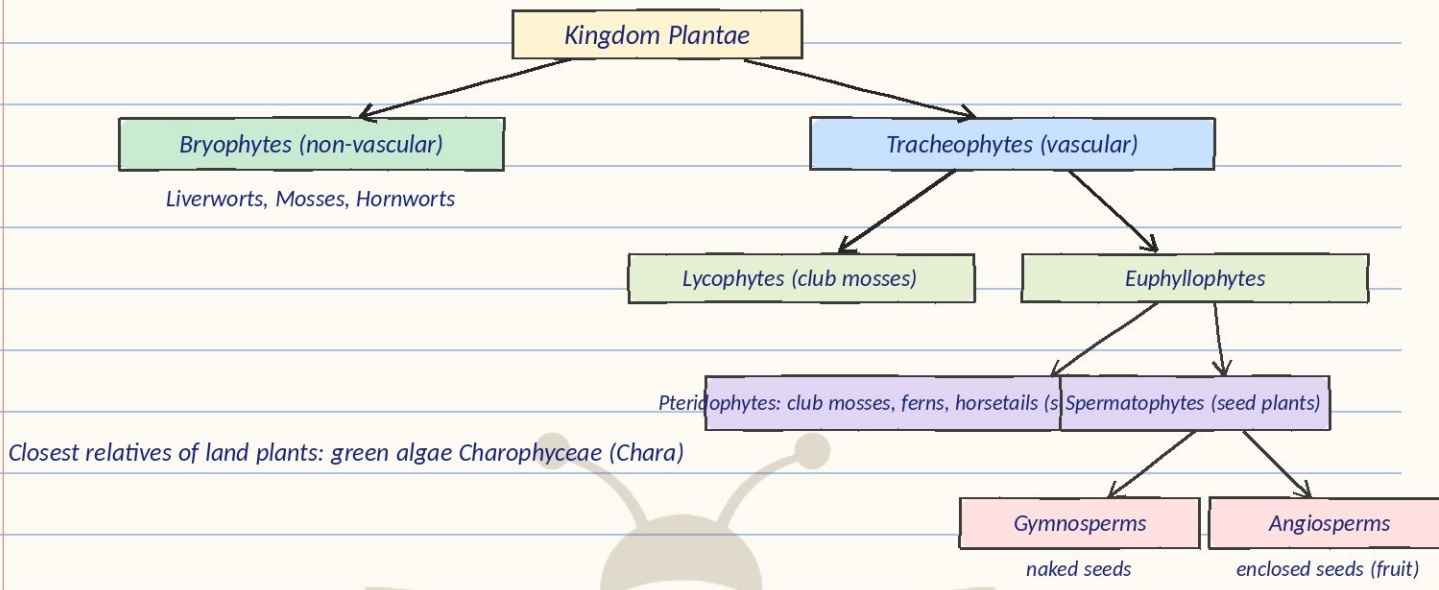



*Fig: Phylogeny of Kingdom Plantae*

## Q5. [SLO 5] Explain the general characteristics of pteridophytes.

**Ans:**

| Class | Features |
|---|---|
| Psilopsida | Whisk ferns; rhizomes, no true roots / leaves; Psilotum |
| Lycopsida | Club mosses; microphylls; strobili; Lycopodium, Selaginella |
| Sphenopsida | Horsetails; jointed hollow stems; Equisetum |
| Fillicinae | Ferns; fronds with sori; circinate vernation; independent sporophyte |

- Importance: ornamentals, edible, drugs, building, coal formation.

## Q6. [SLO 6] Explain the life cycle of fern (Adiantum).

**Ans:**

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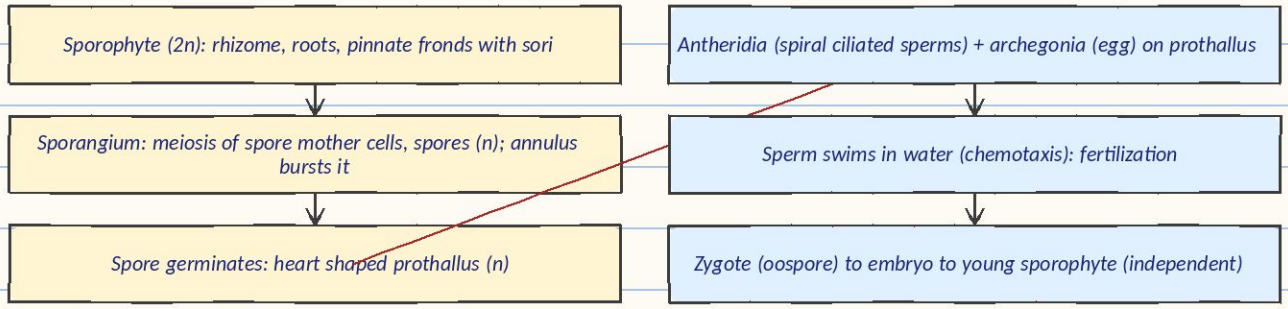


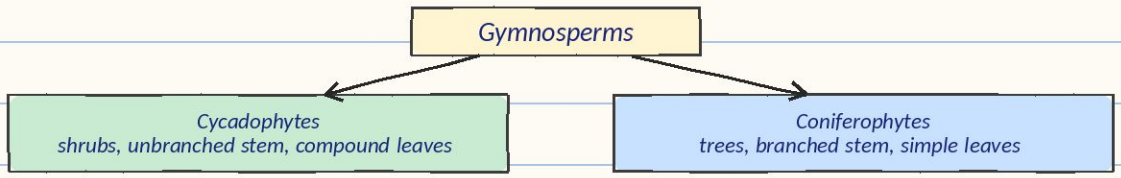

*Fig: Life cycle of fern (Adiantum)*

- *Sporophyte: rhizome, ramenta, adventitious roots, pinnate fronds with sori. Sporangium with annulus and stomium releases spores (meiosis).*
- *Gametophyte: heart shaped prothallus with rhizoids, archegonia and antheridia; antherozoids swim by chemotaxis; zygote to embryo to sporophyte; heteromorphic alternation.*

***Q7. [SLO 7] Describe the general characteristics of gymnosperms and classify them.***

***Ans:***

- *Naked seeded, woody trees and shrubs; heterosporous; seeds without fruits; xerophytic leaves; tracheids only; no companion cells; cones; polyembryony; dominant sporophyte.*




*Orders: Cycadofilicales (extinct), Bennettitales (extinct), Cycadales (living)Cordaitales (extinct), Ginkgoales, Coniferales, Gnetales (living)*

*Genera: Cycas (sago palm), Pinus, Taxus (yew), Picea, Cedrus (deodar), Ginkgo*

*Fig: Classification of gymnosperms*

- *Importance: Ephedra, softwood, paper, sago, resins and turpentine.*

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## Q8. [SLO 8] Describe the life cycle of Pinus.

**Ans:**

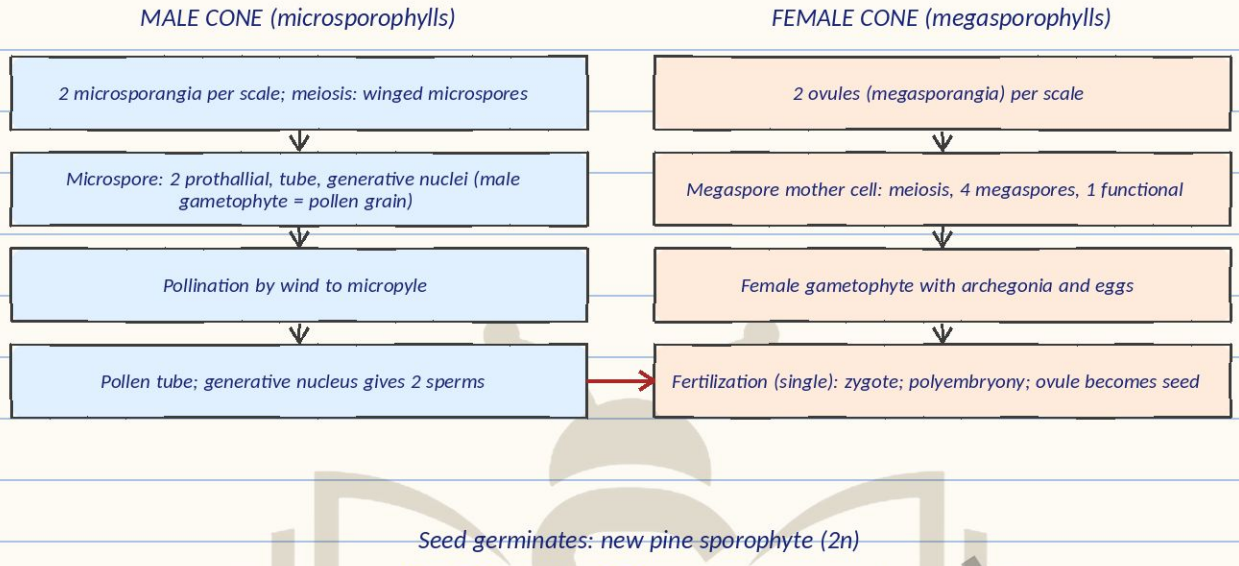



*Fig: Life cycle of Pinus (gymnosperm)*

- *Sporophyte: tall woody tree, needles, taproot; male cones with microsporophylls; female cones with megasporophylls and ovules.*
- *Pollination by winged pollen grains; endosporic male gametophyte (prothallial, tube, generative nuclei, 2 sperms); female gametophyte with archegonia; single fertilization; polyembryony; ovule becomes seed.*

## Q9. [SLO 9] Describe the general characteristics of angiosperms and classify them.

**Ans:**

- *Flowering plants; seeds enclosed in fruit; vessels, fibres and companion cells; dominant sporophyte with flower; double fertilization.*

| Dicots | Monocots |
|---|---|
| Two cotyledons, reticulate venation | One cotyledon, parallel venation |
| Tap root; vascular bundles in ring; cambium | Fibrous roots; scattered bundles; no cambium |
| Floral parts in 4 or 5 | Floral parts in 3 |
| Rose, pea | Wheat, rice, lily |

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## Q10. [SLO 10] Explain the life cycle of an angiosperm.

**Ans:**

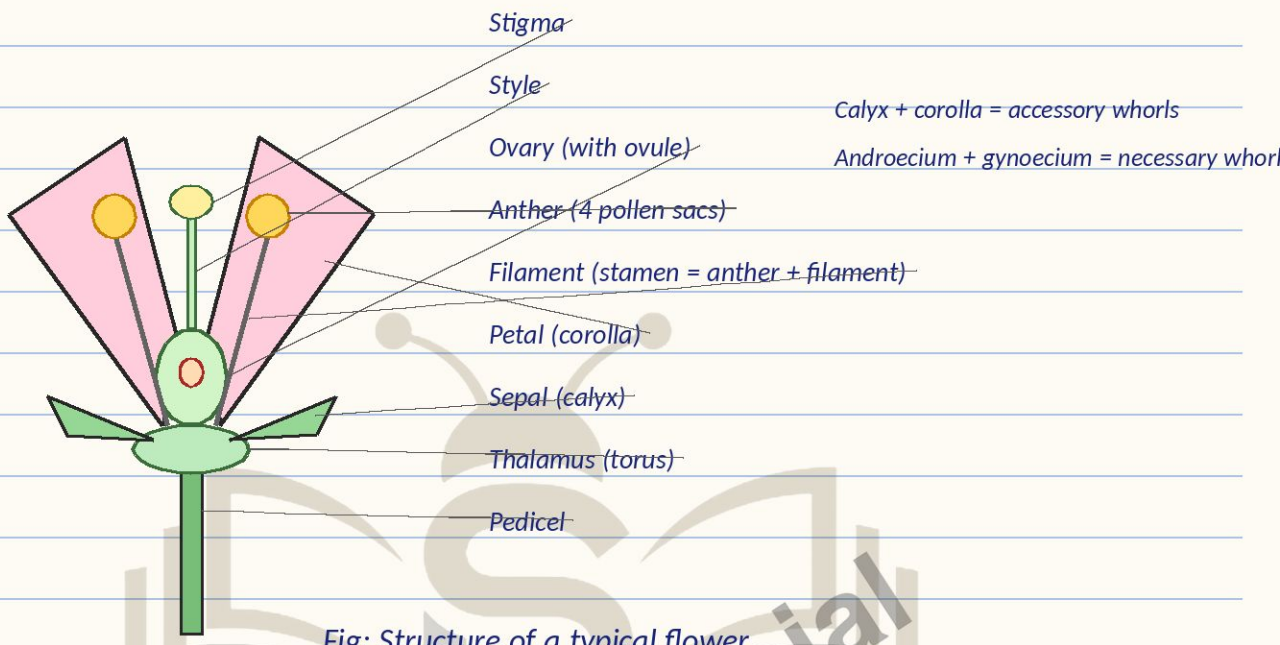



Fig: Structure of a typical flower

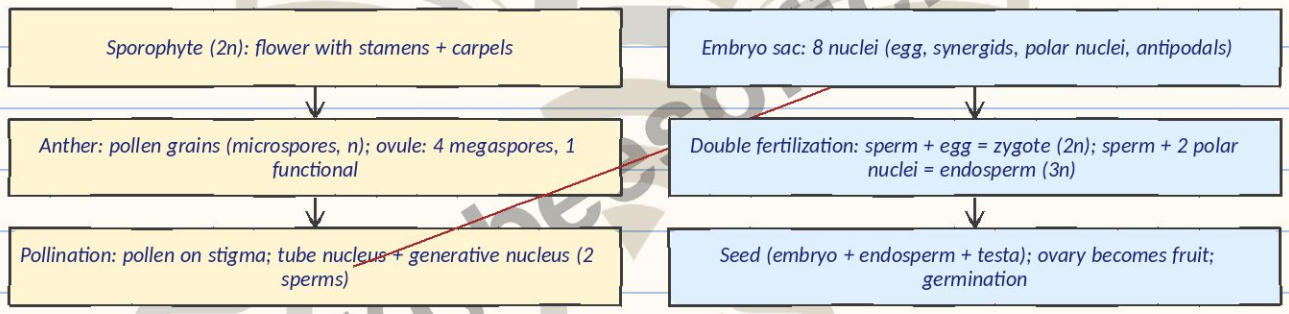



Fig: Life cycle of an angiosperm

- *Sporophyte flower: anther gives pollen (n); ovule gives functional megaspore. Pollination; exosporic male gametophyte (tube nucleus, 2 sperms); embryo sac of 8 nuclei.*
- *Double fertilization: zygote (2n) + triploid endosperm; seed (testa, tegmen); ovary becomes fruit.*

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***Q11. [SLO 11] Describe the general characteristics and economic importance of family Brassicaceae.***

***Ans:***

- *Dicot; mesophytic herbs with pungent smell; stems hairy; leaves simple alternate; bisexual cruciform flowers (4 petals); fruit siliqua or capsule; glucosinolates.*
- *Economic: vegetables, mustard oils, cake as cattle feed and manure, fodder, condiments, medicines (asthma, bronchitis), ornamentals.*

***Q12. [SLO 11] Describe the general characteristics and economic importance of family Solanaceae.***

***Ans:***

- *Dicot; herbs, shrubs; hairy or prickly stem; alternate leaves with odour; pentamerous showy flowers; berry or capsule; alkaloids.*
- *Economic: potato, tomato, brinjal, chillies, ground cherry; tobacco; drugs atropine and daturine; ornamentals (Petunia).*

***Q13. [SLO 11] Describe the general characteristics and economic importance of family Poaceae.***

***Ans:***

- *Monocot; grasses with culms; linear leaves with parallel venation; reduced wind pollinated flowers; fruit caryopsis.*
- *Economic: cereals (wheat, rice, maize, barley, oats, rye), sugarcane, fodder, bamboo, lawn grasses, lemon grass oil, paper, alcohol, fibres.*

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